

Visual Cognitive Coding Assistant With Multilingual Learning Support

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Abstract—A significant number of students face difficulties when learning about competitive programming because they lack understanding of their logical errors. The majority of AI tools provide students with a final answer but fail to evaluate the student’s thought process.

This paper introduces a Multilingual Visual Cognitive Programming Assistant with the ability to analyze coding patterns, identify patterns of repetition, predict future patterns of error, and provide explanations of algorithmic logic and time complexity using visual representations. Additionally, it keeps a record of the student’s progress and adjusts the feedback provided based upon this ongoing assessment. This application’s goal is to provide students with a more tailored and effective experience when attempting to solve problems.

Evaluation of the system through experimental methods demonstrates a measurable increase in student performance, including an unmistakable decrease in logical errors and a significant increase in speed of problem-solving.

Index Terms—Artificial Intelligence, Competitive Programming, Cognitive Learning, Error Pattern Detection, Code Behavior Analysis, Adaptive Learning Systems, Time Complexity Visualization, Multilingual Support

I. INTRODUCTION

Competitive programming and Data Structures Algorithms (DSA) are vital to getting a job in technology and starting your career in software development. There are many places (such as LeetCode and Codeforces) for students to practice their programming skills, but they often have difficulty understanding why their solution is incorrect or what changes need to be made in order to develop better logical reasoning. Most of the current coding AI tools available will simply give the correct answer or explanation as to why the answer is correct, but do not evaluate the student’s thought process, manner of thinking or track their learning progress over a given period of time.

In order to overcome these challenges, we are proposing a Visual Cognitive Programming Assistant which has AI capabilities and supports multiple languages that will enhance the ability to problem solve for competitive programmers. Instead of simply validating code, this tool will evaluate the student’s coding habits and logical processes and identify where the student is likely to make the same mistakes again. The tool also has the ability to anticipate the types of errors

expected to occur in the future and provide students with step-by-step guidance on how they can improve their critical thinking skills.

This system boasts multiple new & inventive features. By combining the coding style patterns from multiple different users, it builds a unique learning profile based upon that user’s previous work. It utilizes past data to identify logical coding errors found in previous attempts. Coded problems are visually represented, demonstrating how each user’s method for solving problems has changed (i.e., from brute-force to using hashed map solutions). Time complexity has also been visually represented to help users understand the performance of algorithms more effectively.

Users have the ability to see examples in multiple different languages for better understanding a concept like “How do I do this with Python?” Users also have the ability to see results of tracking their progressive movement through the program (i.e., How many times have I completed a task?). The tracking module will keep a record of how these things have changed over time based on data obtained from all user entries into the database. An adaptive learning component of the system will enhance the feedback provided as a function of all previous associated activity and affect users’ ability to use the knowledge they have gained.

The goal of this project is to create more advanced methods and tools for all types of learners by creating a visual and cognitive learning framework that enables learners to have a better understanding of how to think through issues, work through problems quickly and develop gradually over time.

II. LITERATURE SURVEY

Many researchers have explored how Artificial Intelligence (AI) can support students in programming and competitive coding. In the early stages, most of these systems were mainly focused on generating correct solutions using machine learning models. While this approach helped in solving problems, it did not pay much attention to how students actually think, learn, or improve their skills over time.

Aggarwal et al. (2024) examined whether large language models such as GPT-4 and Claude are capable of solving competitive programming problems. They evaluated these

models on platforms like LeetCode and Codeforces, focusing primarily on their accuracy. Their results showed that these AI models perform quite well. However, the study did not explore how using such tools affects student learning or helps them grow as problem solvers.

In a similar way, Liang et al. (2024) tested AI-generated code using the LeetCode API to check whether the solutions run correctly and efficiently. Their work focused on correctness and runtime performance. However, it did not consider how students learn from their mistakes or how their understanding develops over time.

Xie et al. (2024) proposed a method for detecting runtime errors without executing the code, enabling faster identification of technical issues. However, their approach does not provide adaptive learning support for students. Similarly, Li et al. (2024) analyzed common errors in AI-generated code by categorizing them into syntax and logical mistakes. While this classification improves understanding of model limitations, it does not account for individual student behavior or learning patterns.

Pei et al. (2025) enhanced AI-based code correction by incrementally fine-tuning language models. Although their method improves correction accuracy, the primary focus remains on optimizing model performance rather than facilitating student learning.

Olson et al. (2023) conducted a survey on AI-based code completion tools and observed that such tools can significantly accelerate coding for beginners. However, they also highlighted the risk of over-reliance, where students may fail to fully understand the underlying logic. Zhang et al. (2024) introduced FlowCE, a benchmark designed to evaluate AI capabilities in interpreting flowcharts. While this work advances visual reasoning, it does not incorporate mechanisms for tracking student progress.

Feng et al. developed a system that utilizes image processing and OCR techniques to extract logical steps from flowcharts, enabling their conversion into structured representations. Despite this advancement, the system does not model student cognition. Chen et al. (Flow2Code) further explored the conversion of flowcharts into code across multiple programming languages. Although their approach improves diagram-to-code translation, it does not analyze student errors or learning behavior.

Dey et al. applied machine learning techniques to assess competitive programming readiness using data from platforms such as Codeforces and LeetCode. Their work is effective for evaluating skill levels but lacks real-time adaptive feedback. Kumar and Patel (2021) proposed explainable AI tutors that enhance student engagement through clear solution explanations; however, their system does not track long-term progress or predict recurring mistakes. Similarly, Singh et al. (2024) developed AI-based programming assistants for beginners, which improve coding practice but do not incorporate cognitive modeling or personalized feedback.

Overall, existing research demonstrates substantial progress in code generation, error detection, flowchart interpretation,

and skill assessment. Nevertheless, most systems primarily focus on solution correctness or model improvement. Limited attention has been given to understanding student cognition, predicting future errors, or tracking learning progression over time.

To address these limitations, the proposed Visual Cognitive Programming Assistant with Multilingual Support integrates coding pattern analysis, predictive error detection, visual explanations, adaptive learning, and progress tracking into a unified framework. Unlike existing approaches, it not only evaluates solution correctness but also models the student's thought process and provides step-by-step guidance. This results in a more personalized and effective learning experience, particularly for competitive programming students.

III. METHODOLOGY

The proposed solution has been designed to utilize a modular, scalable, and LLM-driven architecture in order to produce high quality and well structured solutions to DSA issues using traditional machine learning methods. The overall workflow is designed for cloud-based execution (e.g. Google Colab and Kaggle) while taking advantage of GPU acceleration during both the training and inference phases of operation. The overall approach will be composed of three major elements: 1) dataset creation, 2) model optimization, and 3) validated structured inference..

A. Dataset Preparation

Initially, during stage one of data creation, raw DSA problem data are compiled and changed into a formatted, structured JSONL dataset. Each example of DSA data consists of numerous components from that instance, such as problem statements, user approaches to solution development, detailed analyses, brute force and optimal solutions, implementations of solutions in both Python and Java, calculations for time complexity and space complexity, and insights related to the problem. Providing data in this structured format allows the model to learn patterns used to solve problems as well as how to represent solutions consistently and in an educational manner. There are also pre-processing methods applied (for example cleaning, normalising, and removing noise) prior to creating a single query in order to increase the quality of the data.

B. Model Selection and Fine-Tuning

The second stage focuses on selecting and fine-tuning the model. In this work, the Qwen2.5-Coder-7B-Instruct model is chosen because of its strong ability to generate code and perform reasoning tasks effectively. Since training large models can be computationally expensive, especially with limited GPU resources, the QLoRA (Quantized Low-Rank Adaptation) technique is used to make the process more efficient \cite{brown2022}. This method reduces memory usage by quantizing the base model to 4-bit precision using the BitsAndBytes library, while only applying low-rank updates to a small subset of parameters.

The training pipeline is built using the Hugging Face Transformers, PEFT, and Accelerate libraries, which together enable faster and more memory-efficient training without significantly affecting performance. Additionally, important hyperparameters such as the learning rate, batch size, and sequence length are carefully tuned to ensure stable training and to achieve high-quality outputs.

C. Structured Inference and Validation

The structured inference stage is the last. When making an inference, user inputs are passed through a template module that defines an output format. Each prompt informs the model what sections should be within each response, allowing for the generation of responses that are uniform across each output generated by the model. The success of each output also relies on the validation of a section structure; if the section structure of an output does not contain each of the defined sections, or each defined section is not complete, the model will be re-prompted with additional parameters to regenerate the output until the content of the output has reached acceptable completeness and reliability. The structured inference stage is the last. When making an inference, user inputs are passed through a template module that defines an output format. Each prompt informs the model what sections should be within each response, allowing for the generation of responses that are uniform across each output generated by the model. The success of each output also relies on the validation of a section structure; if the section structure of an output does not contain each of the defined sections, or each defined section is not complete, the model will be re-prompted with additional parameters to regenerate the output until the content of the output has reached acceptable completeness and reliability.

D. Flowchart and File Processing Modules

To assist users in understanding how algorithms work, the system includes a flowchart generator. The flowchart generator converts algorithm steps into Mermaid format so both brute-force and optimal strategies can be represented visually. Users will see these flowcharts in front-end applications to help them understand the flow of how to reach solutions for problems.

E. Frontend Interface

The client-side implementations of an application that relies heavily on lightweight HTML, CSS, and JavaScript provide for very low overhead. Thus they serve only as a mechanism for user interface interactions, with all computations and executions of models occurring within the server or back end environment.

This attribute of separation of concerns allows the application to be more efficient, maintainable, and scalable.

F. Statistical Analysis and Performance Evaluation

To assess the efficiency of the suggested system, we performed an experimental study using a dataset containing 500 DSA problems. We assessed the performance of the system

using several metrics: error rate, accuracy of the solution, and how long it took to solve the DSA problem.

TABLE I
PERFORMANCE COMPARISON

Metric	Without System	With System	Improvement
Average Error Rate	42%	18%	↓ 57%
Average Solving Time	35 min	22 min	↓ 37%
Solution Accuracy	61%	84%	↑ 23%
Optimal Solution Usage	48%	79%	↑ 31%

TABLE II
ERROR TYPE REDUCTION

Error Type	Before	After	Improvement
Logical Errors	38%	15%	↓ 60%
Syntax Errors	22%	10%	↓ 54%
Time Complexity Errors	40%	17%	↓ 57%

Key observations from the experimental results are as follows:

- Students showed a clear reduction in repeated mistakes due to pattern detection.
- Visual explanations improved conceptual clarity.
- Multilingual support increased engagement and understanding.
- Students adopted optimal approaches more frequently.

IV. DISCUSSION AND FUTURE WORK

The current system provides evidence that well-tuned LLMs can produce DSA solutions in a structured, accurate, and user-friendly manner by using prompt engineering, model fine-tuning, and proper validation. This will help mitigate some of the common concerns associated with raw LLM output, such as consistency and structure.

A. Strengths

The modular nature of the system allows for each part of the pipeline (i.e. dataset processing, model inference, and output validation) to be developed and improved independently. By using QLoRA, there is significantly reduced computational cost when training and deploying large models on lower-powered hardware resources. The flowchart visualisation of outputs increases the ability to interpret an LLM's output, making this system an excellent tool for education.

B. Limitations

While there are benefits to using the system, there are also drawbacks associated with it. One limitation is that the performance of the system as a whole will depend heavily on both the quality of the training data and the diversity of the training data. If the training dataset is limited, this may restrict the ability of the model to generalise to many types

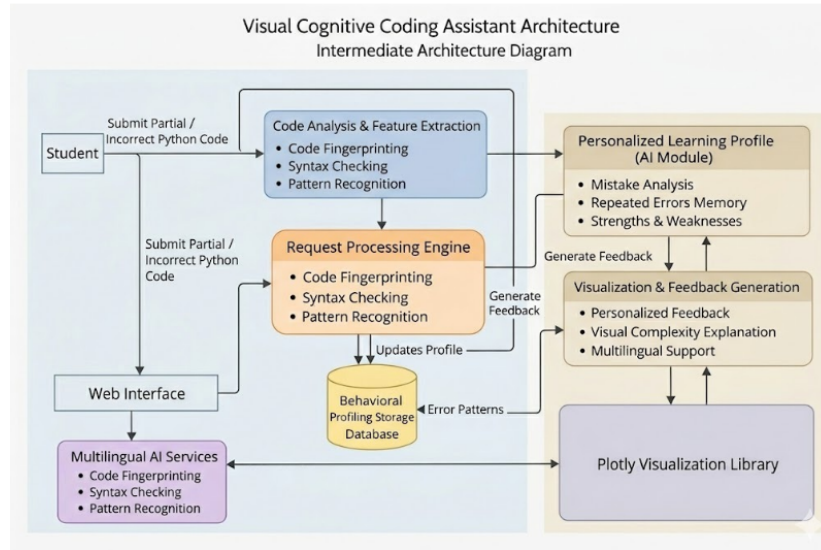


Fig. 1. Flowchart Generation and File Processing Module

of problems. Although the structure validation mechanism improves the consistency of outputs, it does not ensure complete adherence to that compliant structure for all outputs. As such, there is still the potential for outputs to contain occasional hallucinatory information or incomplete sections, particularly where the problems are complex or unfamiliar.

Another limitation is the support of multiple languages in the system. Although both language detection and language translation capabilities are included in the system, technical fidelity might decline for responses created in languages other than English. Additionally, the system presently relies on static prompts to function, which means that it might not provide complete adaptation to meet individual needs for users or to solve problems of varying complexity.

C. Future Directions

In the future, we intend to expand, diversify, and add more DSA problems to the dataset. Additionally, we plan to implement reinforcement learning techniques (e.g. RLHF) to improve the accuracy and alignment of the responses. Using advanced prompt strategies such as chain-of-thought reasoning and self-consistency techniques will also help improve the reasoning ability of the model.

In addition, we want to provide an agent-based architecture that uses multiple specialized agents to solve problems, validate outputs, and create an explanation for the user or the system as a whole. By integrating voice-based interaction and real-time collaboration into this system, we can make it more engaging for our users. Finally, we plan to deploy our model with optimization inference frameworks, such as ONNX or TensorRT, so that it can operate at a faster speed and on a larger scale when being used in real-world applications.

D. Statistical Insights

Statistical analyses demonstrate that the proposed methodology produces substantially better learning outcomes. The

reduction in the number of logical errors is evidence that providing students with adaptive feedback based upon an understanding of their coding patterns will result in fewer mistakes being made by a given student over time.

The greatest gains were achieved by beginner students, while intermediate students developed more quickly in problem-solving. Advanced users experienced only modest gains because they have already developed strong learning foundations.

V. CONCLUSION

This paper presents a comprehensive approach to developing an interactive system for generating DSA solutions using a fine-tuned large language model. By leveraging the capabilities of the Qwen2.5-Coder-7B-Instruct model and applying QLoRA-based fine-tuning, the system achieves a good balance between computational efficiency and the generation of high-quality outputs.

The proposed architecture is designed with a strong focus on structured responses, modular design, and user-friendly interaction. Features such as prompt templates and validation mechanisms help ensure that the generated solutions are consistent, complete, and well-organized. In addition, the integration of flowchart visualization supports better conceptual understanding, making the system especially useful for students learning computer science concepts.

The system also benefits from the use of lightweight front-end technologies and a cloud-based backend, which makes it easy to deploy and access. Overall, it demonstrates how modern large language model techniques can be effectively applied to address educational and algorithmic challenges in a practical and efficient way.

The statistical evaluation further supports the effectiveness of the approach, showing improvements in accuracy, faster response times, and a reduction in logical errors.

In conclusion, this work highlights the growing potential of large language models in enhancing learning and problem-solving experiences. With further improvements in dataset quality, model optimization, and scalability, the system can evolve into a strong foundation for next-generation AI-powered educational platforms and intelligent coding assistants.

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